

24
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2015

Magnetism And Electromagnetism

Magnetic field Intensity, H

The field intensity (magnetic field strength) or magnetising force, H at any point in magnetic field is the force experience by a unit north pole (1 weber) placed at that point.

$$H = \frac{4\pi \times 10^3}{r^2}$$

The magnetic flux density is the flux per unit area at right angle to the flux

$$B = \frac{\Phi}{A} \text{ wb/m}^2$$

Relation Between Magnetising force, H and flux density, B

$$B = \mu_r \mu_0 H$$

Magnetomotive force, mmf

Magnetomotive force, mmf is the amount of current turns.

$$\text{mmf} = NI \text{ AT}$$

or magnetomotive force

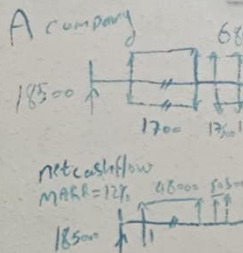
Reluctance:

It is the magnetomotive force per unit flux, and denoted, S

$$S = \frac{\text{mmf}}{\Phi} \text{ AT/wb}$$

$$S = \frac{IN}{\Phi}$$

$$S = \frac{l}{\mu_r \mu_0 A}$$



Magnetism And Electromagnetism

Example (1):

A circular ring of iron is wound uniformly with 1000 turns of wire. Its mean diameter is 20 cm and its cross-sectional area 4 cm^2 . Determine the current when will produce a flux of $2 \times 10^{-5} \text{ wb}$.

Assuming the permeability to be 800

Solution

$$N = 1000 \text{ turns} \quad d = 20 \text{ cm} = 20 \times 10^{-2} \text{ m} = 0.20 \text{ m}$$

$$A = 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2 \quad I = ?$$

$$\Phi = 2 \times 10^{-5} \text{ wb}$$

$$l = 2\pi r = d\pi = 0.20\pi$$

$$S = \frac{\text{mmf}}{\Phi}$$

$$\frac{l}{\mu_r \mu_0} = \frac{NI}{\Phi}$$

$$I = \frac{\Phi l}{\mu_r \mu_0 N}$$

$$= \frac{2 \times 10^{-5} \times 0.20\pi}{800 \times 4\pi \times 10^{-7} \times 4 \times 10^{-4} \times 1000}$$

$$I = 0.03125$$

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Magnetism And Electromagnetism

Example (1):

An iron ring of 50 cm mean diameter 10 cm^2 cross-section has 1000 turns of insulated copper wire wound uniformly on it. If it produces a flux of $2.5 \times 10^{-3} \text{ wb}$ when a current of 4 ampere flows in it. Calculate the relative permeability of the iron.

Solution

diameter, $d = 50 \text{ cm} = 50 \times 10^{-2} \text{ m} = 0.5 \text{ m}$
 Area, $A = 10 \text{ cm}^2 = 10 \times 10^{-4} \text{ m}^2$ $N = 1000$ turns
 flux, $\phi = 2.5 \times 10^{-3} \text{ wb}$
 current, $I = 4 \text{ A}$

relative permeability $\mu_r = ?$

$$B = \mu_r \mu_0 H$$

$$\mu_r = \frac{B}{\mu_0 H}$$

$$B = \frac{\phi}{A} = \frac{2.5 \times 10^{-3}}{10 \times 10^{-4}}$$

$$B = 2.5 \text{ wb/m}^2$$

$$H = \frac{NI}{L} = \frac{NI}{\pi d} = \frac{1000 \times 4}{\pi (0.5)} = \frac{8000}{\pi} \text{ AT/m}$$

$$L = \text{circumference} = 2\pi r = \pi d$$

$$\mu_r = \frac{2.5}{\frac{8000}{\pi} \times \frac{1}{\pi}} = \frac{2.5}{32 \times 10^{-5}}$$

$$\mu_r = 7812.5$$

Example (2):

It is desired to establish a flux of $63 \times 10^{-5} \text{ wb}$ in closed iron circuit of 24 cm mean length and 4 cm^2 cross-section.

Assuming μ_r for iron to 900, Calculate the ampere turns required.

Solution $\phi = 63 \times 10^{-5} \text{ wb}$

$$L = 24 \text{ cm} = 24 \times 10^{-2} \text{ m}$$

$$A = 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$$

$$\mu_r = 900$$

$$\text{ampere-turns} = NI = m.m.f$$

$$m.m.f = NI$$

$$S = \frac{m.m.f}{\phi}$$

$$m.m.f = \phi \cdot S = \phi \cdot \frac{L}{\mu_r \mu_0 A}$$

$$m.m.f = \frac{63 \times 10^{-5} \times 24 \times 10^{-2}}{900 \times 4\pi \times 10^{-7} \times 4 \times 10^{-4}}$$

$$m.m.f = 334.25 \text{ AT}$$

Example (4):

In a certain electric motor 1 cm wires that carry a current of 5 A are perpendicular to a magnetic field of 0.8 wb/m^2 .

what is the force on each of these wire.

$$L = 1 \text{ cm} = 1 \times 10^{-2} \text{ m}$$

$$I = 5 \text{ ampere} \quad B = 0.8 \text{ wb/m}^2$$

$$F = ?$$

$$F = B L I$$

$$F = 0.8 \times 1 \times 10^{-2} \times 5$$

$$F = 0.04 \text{ N}$$

Example (5):

A square cross-section magnet has a pole strength of $1 \times 10^{-3} \text{ weber}$ and cross-sectional area $2 \text{ cm} \times 2 \text{ cm}$.

Calculate the field strength of a distance of 10 cm from pole in air.

$$m = 1 \times 10^{-3} \text{ wb}$$

$$\text{area } A = 2 \text{ cm} \times 2 \text{ cm} = 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$$

field strength $H = ?$

$$r = 10 \text{ cm} = 10 \times 10^{-2} \text{ m} = 0.1$$

$$F = \frac{m}{4\pi \mu_r \mu_0 r^2}$$

$$\mu_r = 1$$

$$F = \frac{1 \times 10^{-3}}{4\pi (4\pi \times 10^{-7}) (0.1)^2}$$

$$F = 6332.573 \text{ N}$$

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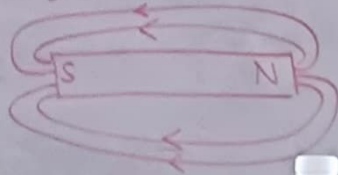
Magnetism And Electromagnetism

Magnetic field:

The field in which a magnetic pole experience a force is called a magnetic field.

Properties of Magnetic lines of Force

- (i) The magnetic line of force forms a closed loop.



- (ii) The direction of the magnetic line is from North pole to the South pole.

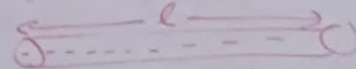
- (iii) Magnetic line of force tends to pass through iron and other magnetic materials.

Note: uniform magnetic field

The magnetic field in a region is uniform if it has the same magnetic flux density and the same direction at all points in the region.

Magnetic Effect of Electric current

Suppose we have a conductor of length l and the current I will flow in this conductor.

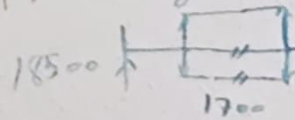


When the current flows in the conductor, it will produce a magnetic field and the force of this magnetism is given

$$F = B l I$$

B : flux density wb/m^2

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Magnetism And Electromagnetism

It was known to the ancient that there exist in nature some materials which possessed the property of attracting and holding small piece of iron.

Magnets And Magnetism

Artificial magnet may be classified into two general class:

1. Permanent magnets
2. Electromagnets.

Permanent magnets have the property of retaining their magnetism indefinitely and they require no permanent electric aid to magnetism, like exciting coil (ampere-turns).

Electromagnets on the other hand have their magnetism due to flow of an electric current.

Laws of Magnetic Force

- (i) Like poles repel each other while unlike poles attract each other.

(ii) The force between two magnetic poles is directly proportional to the product of their pole strength and inversely proportional to the square distance between them.

$$F \propto m_1 \times m_2$$
$$F \propto \frac{1}{r^2}$$
$$F = k \frac{m_1 m_2}{r^2}$$

$$k = \frac{1}{\mu_r \mu_0}$$

μ_r : Permeability for materials

$$\mu_0 = 4\pi \times 10^{-7} \text{ Wb/Am}$$

